Section 2: Mechanics II — Solutions

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1. Gravitational Dependence

For a simple pendulum,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

For fixed length L, the period is proportional to $1/\sqrt{g}$.

Period on the Moon

Given $T_E = 4$ s on Earth and $g_M = g_E/6$ on the Moon,

$$\frac{T_M}{T_E} = \sqrt{\frac{g_E}{g_M}} = \sqrt{\frac{g_E}{g_E/6}} = \sqrt{6}$$

Thus,

$$T_M = 4\sqrt{6} \approx 9.80 \text{ s}$$

Length for a 1-second pendulum on Earth

Set $T = 1$ s and solve for L:

$$L = g\left(\frac{T}{2\pi}\right)^2$$

Using $g = 9.81$ m/s^2,

$$L = 9.81\left(\frac{1}{2\pi}\right)^2 \approx 0.248 \text{ m}$$

So,

$$\boxed{T_{\text{Moon}} \approx 9.80 \text{ s}, \qquad L \approx 0.248 \text{ m}}$$

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2. Harmonic Motion

Given

$$x(t) = 0.2\cos(10\pi t)$$

This is simple harmonic motion of the form

$$x(t) = A\cos(\omega t)$$

so

$$A = 0.2 \text{ m}, \qquad \omega = 10\pi \text{ rad/s}$$

For a mass-spring system,

$$\omega = \sqrt{\frac{k}{m}}$$

Hence,

$$k = m\omega^2 = 10(10\pi)^2 = 1000\pi^2 \text{ N/m}$$

Numerically,

$$k \approx 9869.6 \text{ N/m}$$

The total mechanical energy is

$$E = \frac{1}{2}kA^2$$

$$E = \frac{1}{2}(1000\pi^2)(0.2)^2 = 20\pi^2 \text{ J}$$

$$E \approx 197.4 \text{ J}$$

Therefore,

$$\boxed{k = 1000\pi^2 \text{ N/m} \approx 9869.6 \text{ N/m}}$$

$$\boxed{E = 20\pi^2 \text{ J} \approx 197.4 \text{ J}}$$

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3. Conservation of Energy

The pendulum is released from rest at angle $\theta = 15^\circ$ with length $L = 1.0$ m.

The loss of gravitational potential energy becomes kinetic energy at the bottom:

$$mgL(1 - \cos\theta) = \frac{1}{2}mv^2$$

Mass cancels, so

$$v = \sqrt{2gL(1 - \cos\theta)}$$

Substitute values:

$$v = \sqrt{2(9.81)(1.0)(1 - \cos 15^\circ)} \approx 0.818 \text{ m/s}$$

So,

$$\boxed{v \approx 0.818 \text{ m/s}}$$

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4. Energy & Momentum

Speed of the 0.5 kg block at the bottom

From energy conservation,

$$mgh = \frac{1}{2}mv^2$$

Thus,

$$v = \sqrt{2gh} = \sqrt{2(9.81)(3.0)} \approx 7.67 \text{ m/s}$$

Perfectly inelastic collision

The block sticks to a 1.5 kg block at rest, so linear momentum is conserved:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2)V$$

With $m_1 = 0.5$, $v_1 = 7.67$, $m_2 = 1.5$, $v_2 = 0$,

$$V = \frac{0.5 \cdot 7.67}{0.5 + 1.5} \approx 1.92 \text{ m/s}$$

Therefore,

$$\boxed{V \approx 1.92 \text{ m/s}}$$

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5. Inelastic Collision

Use conservation of momentum:

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2)V$$

Given

$$m_1 = 70 \text{ kg}, \quad v_1 = 3 \text{ m/s}, \quad m_2 = 140 \text{ kg}, \quad v_2 = 0$$

Then

$$V = \frac{70 \cdot 3}{70 + 140} = \frac{210}{210} = 1 \text{ m/s}$$

So,

$$\boxed{V = 1 \text{ m/s}}$$

Is kinetic energy conserved?

Initial kinetic energy:

$$K_i = \frac{1}{2}(70)(3^2) = 315 \text{ J}$$

Final kinetic energy:

$$K_f = \frac{1}{2}(210)(1^2) = 105 \text{ J}$$

Since $K_f \neq K_i$, kinetic energy is not conserved.

$$\boxed{\text{No, kinetic energy is not conserved.}}$$

The missing mechanical energy is transformed into heat, sound, and deformation during the inelastic collision.

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6. Energy Dissipation

If the ball loses 30% of its mechanical energy after each bounce, then it keeps 70% of its energy.

Since gravitational potential energy is proportional to height,

$$h_{n+1} = 0.7h_n$$

Starting from $h_0 = 2.0$ m:

After the first bounce,

$$h_1 = 0.7(2.0) = 1.4 \text{ m}$$

After the second bounce,

$$h_2 = 0.7(1.4) = 0.98 \text{ m}$$

Therefore,

$$\boxed{h_2 = 0.98 \text{ m}}$$

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7. Dynamics with Friction

We consider the lower 10 kg block.

Friction from the top block

The 5 kg block is tied to the wall, so the lower block slides under it. The kinetic friction magnitude between the blocks is

$$f_{12} = \mu_k N_{12} = 0.2(5g) = 0.2(5)(9.81) = 9.81 \text{ N}$$

This acts opposite the motion of the lower block.

Friction with the ground

The lower block supports both masses, so the normal force from the ground is

$$N_g = (10 + 5)g = 15g$$

Hence the ground friction is

$$f_g = \mu_k N_g = 0.2(15)(9.81) = 29.43 \text{ N}$$

Net force on the 10 kg block

$$F_{\text{net}} = 45 - 9.81 - 29.43 = 5.76 \text{ N}$$

Thus,

$$a = \frac{F_{\text{net}}}{10} = \frac{5.76}{10} = 0.576 \text{ m/s}^2$$

So,

$$\boxed{a \approx 0.576 \text{ m/s}^2}$$

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8. Work of a Variable Force

Given

$$F(x) = -kx$$

Equation of motion and solution

Newton's second law gives

$$m\frac{d^2x}{dt^2} = -kx$$

or

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$$

This is the equation of simple harmonic motion. Its general solution is

$$x(t) = A\cos(\omega t) + B\sin(\omega t)$$

where

$$\omega = \sqrt{\frac{k}{m}}$$

Work done from 0 to x_0

The work done by the force is

$$W = \int_0^{x_0} F(x)\,dx = \int_0^{x_0} (-kx)\,dx$$

$$W = -k\left[\frac{x^2}{2}\right]_0^{x_0} = -\frac{1}{2}kx_0^2$$

Thus,

$$\boxed{W = -\frac{1}{2}kx_0^2}$$

Interpretation as potential energy

For a conservative force,

$$W = -\Delta U$$

Choosing $U(0) = 0$, we obtain

$$U(x) = \frac{1}{2}kx^2$$

So,

$$\boxed{U(x) = \frac{1}{2}kx^2}$$

Verify $F = -\frac{dU}{dx}$

Differentiate $U(x)$:

$$\frac{dU}{dx} = kx$$

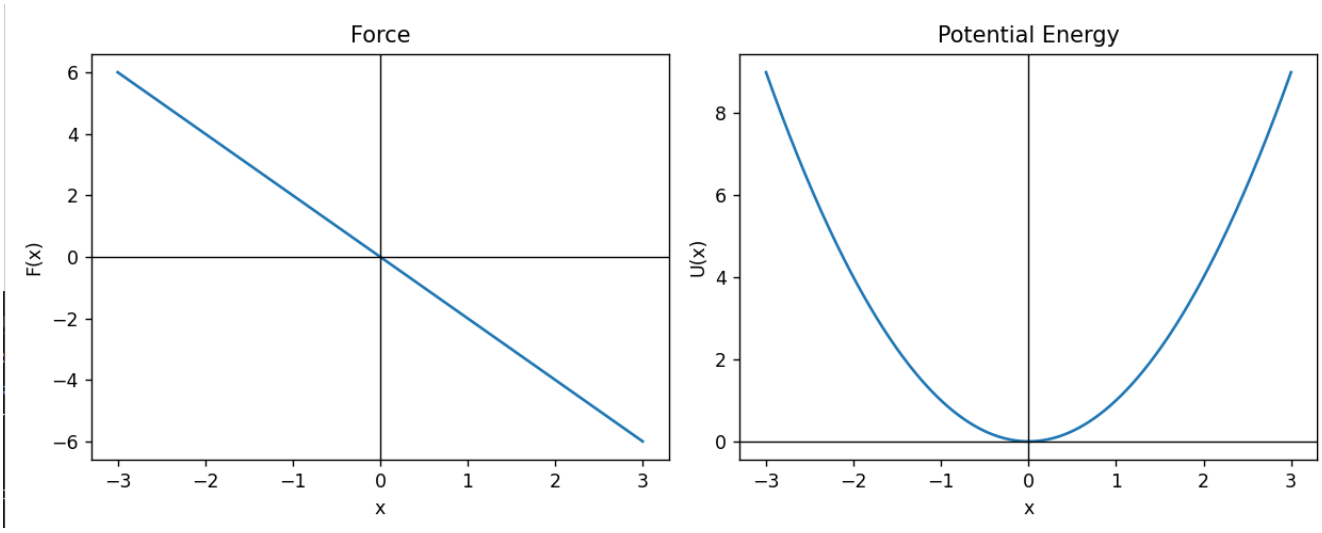
Therefore,

$$F = -\frac{dU}{dx} = -kx$$

which matches the given force.

Graphs of $F(x)$ and $U(x)$

- $F(x) = -kx$ is a straight line through the origin with negative slope.
- $U(x) = \frac{1}{2}kx^2$ is an upward-opening parabola.



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9. Vertical Throw with Drag

We solve

$$m\frac{dv}{dt} = -mg - kv$$

with $v(0) = v_0$ and $x(0) = 10$.

Analytical solution for velocity

Rewrite as

$$\frac{dv}{dt} + \frac{k}{m}v = -g$$

This linear differential equation has solution

$$v(t) = \left(v_0 + \frac{mg}{k}\right)e^{-kt/m} - \frac{mg}{k}$$

Position as a function of time

Integrating $v(t)$,

$$x(t) = 10 + \frac{m}{k}\left(v_0 + \frac{mg}{k}\right)(1 - e^{-kt/m}) - \frac{mg}{k}t$$

Maximum height

At the highest point, $v(t_*) = 0$:

$$\left(v_0 + \frac{mg}{k}\right)e^{-kt_*/m} - \frac{mg}{k} = 0$$

So,

$$t_* = \frac{m}{k}\ln\left(1 + \frac{kv_0}{mg}\right)$$

Substitute into $x(t)$:

$$x_{\max} = 10 + \frac{mv_0}{k} - \frac{m^2 g}{k^2} \ln\left(1 + \frac{kv_0}{mg}\right)$$

Comparison with no drag

If $k = 0$, then

$$v(t) = v_0 - gt$$

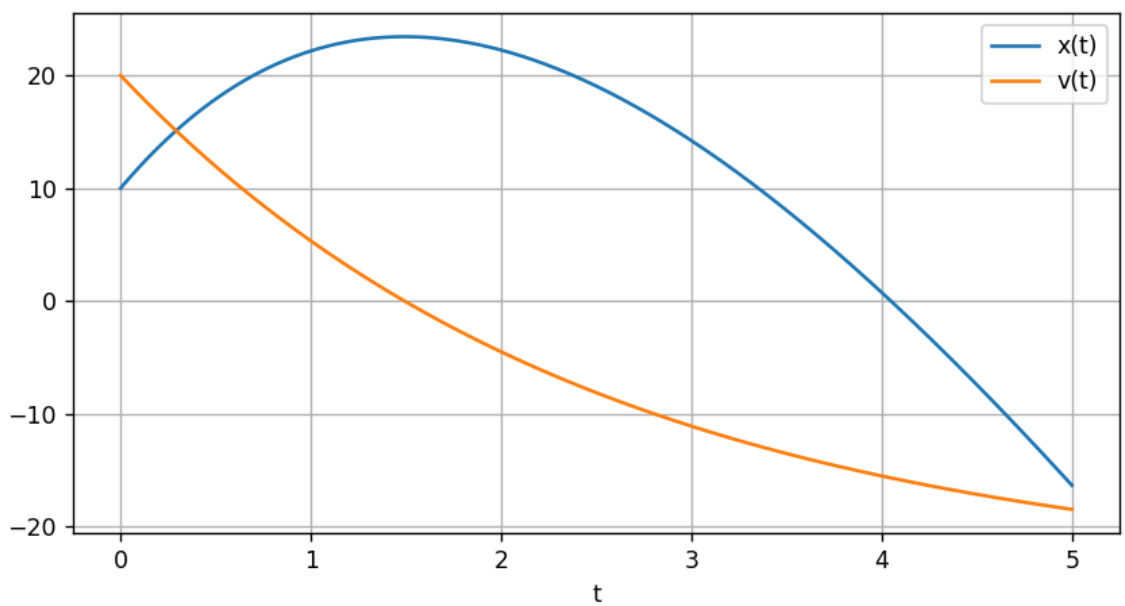
and the maximum height is

$$x_{\max}^{(0)} = 10 + \frac{v_0^2}{2g}$$

Since drag removes energy, we always have

$$x_{\max} < x_{\max}^{(0)}$$

Numerical simulation in Python



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10. Force Field and Power

Given

$$x = 5t^2 - t, \qquad y = 2t^3, \qquad z = -3t + 2$$

and mass $m = 0.5$ kg.

Velocity

$$\vec{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt}\right) = (10t - 1,\ 6t^2,\ -3)$$

Momentum

$$\vec{p}(t) = m\vec{v}(t) = 0.5(10t - 1,\ 6t^2,\ -3)$$

$$\vec{p}(t) = (5t - 0.5,\ 3t^2,\ -1.5)$$

Acceleration

$$\vec{a}(t) = (10,\ 12t,\ 0)$$

Force

$$\vec{F}(t) = m\vec{a}(t) = 0.5(10,\ 12t,\ 0) = (5,\ 6t,\ 0)$$

Power

Power is

$$P(t) = \vec{F} \cdot \vec{v}$$

Thus,

$$P(t) = 5(10t - 1) + (6t)(6t^2) + 0(-3)$$

$$P(t) = 50t - 5 + 36t^3$$

So,

$$\boxed{\vec{v}(t) = (10t - 1,\ 6t^2,\ -3)}$$

$$\boxed{\vec{p}(t) = (5t - 0.5,\ 3t^2,\ -1.5)}$$

$$\boxed{\vec{a}(t) = (10,\ 12t,\ 0)}$$

$$\boxed{\vec{F}(t) = (5,\ 6t,\ 0)}$$

$$\boxed{P(t) = 36t^3 + 50t - 5}$$

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11. Dynamics with a Time-Dependent Force

Given

$$\vec{F}(t) = (15t,\ 3t - 12,\ -6t^2)$$

and mass $m = 3$ kg.

Acceleration

From Newton's second law,

$$\vec{a}(t) = \frac{\vec{F}}{m} = (5t,\ t-4,\ -2t^2)$$

Velocity

Integrate each component and use $\vec{v}(0) = (2,0,1)$:

$$v_x(t) = \int 5t\,dt = \frac{5}{2}t^2 + C_x$$

Since $v_x(0) = 2$, $C_x = 2$, so

$$v_x(t) = \frac{5}{2}t^2 + 2$$

Similarly,

$$v_y(t) = \int (t-4)\,dt = \frac{1}{2}t^2 - 4t + C_y$$

Using $v_y(0) = 0$, $C_y = 0$:

$$v_y(t) = \frac{1}{2}t^2 - 4t$$

And,

$$v_z(t) = \int (-2t^2)\,dt = -\frac{2}{3}t^3 + C_z$$

Using $v_z(0) = 1$, $C_z = 1$:

$$v_z(t) = -\frac{2}{3}t^3 + 1$$

Therefore,

$$\boxed{\vec{v}(t) = \left(\frac{5}{2}t^2 + 2,\ \frac{1}{2}t^2 - 4t,\ -\frac{2}{3}t^3 + 1\right)}$$

Position

Integrate again and use $\vec{r}(0) = (5,2,-3)$.

For x:

$$x(t) = \int \left(\frac{5}{2}t^2 + 2\right) dt = \frac{5}{6}t^3 + 2t + C_1$$

With $x(0) = 5$, $C_1 = 5$:

$$x(t) = \frac{5}{6}t^3 + 2t + 5$$

For y:

$$y(t) = \int \left(\frac{1}{2}t^2 - 4t\right) dt = \frac{1}{6}t^3 - 2t^2 + C_2$$

With $y(0) = 2$, $C_2 = 2$:

$$y(t) = \frac{1}{6}t^3 - 2t^2 + 2$$

For z:

$$z(t) = \int \left(-\frac{2}{3}t^3 + 1\right) dt = -\frac{1}{6}t^4 + t + C_3$$

With $z(0) = -3$, $C_3 = -3$:

$$z(t) = -\frac{1}{6}t^4 + t - 3$$

Thus,

$$\boxed{\vec{r}(t) = \left(\frac{5}{6}t^3 + 2t + 5,\ \frac{1}{6}t^3 - 2t^2 + 2,\ -\frac{1}{6}t^4 + t - 3\right)}$$

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12. Work and Energy with a Constant Force

Given

$$\vec{F} = (6,2)\ \text{N}, \qquad m = 2\ \text{kg}$$

with initial data

$$\vec{v}(0) = (1, -1), \qquad \vec{r}(0) = (0,0)$$

Acceleration

$$\vec{a}(t) = \frac{\vec{F}}{m} = (3,1)$$

So,

$$\boxed{\vec{a}(t) = (3,1)}$$

Velocity

Integrating,

$$\vec{v}(t) = (1, -1) + (3,1)t = (1 + 3t,\ -1 + t)$$

Thus,

$$\boxed{\vec{v}(t) = (1 + 3t,\ -1 + t)}$$

Position

Integrate velocity:

$$x(t) = \int (1 + 3t)dt = t + \frac{3}{2}t^2$$

$$y(t) = \int (-1 + t)dt = -t + \frac{1}{2}t^2$$

Using $\vec{r}(0) = (0,0)$, we get

$$\boxed{\vec{r}(t) = \left(t + \frac{3}{2}t^2,\ -t + \frac{1}{2}t^2\right)}$$

Trajectory

The motion is parabolic in the plane. A parametric plot of $x(t)$,$y(t)$ gives the trajectory.

Work done by the force at $t = 3$ s

The displacement from the origin at $t = 3$ is

$$\vec{r}(3) = \left(3 + \frac{3}{2} \cdot 9,\ -3 + \frac{1}{2} \cdot 9\right) = (16.5,1.5)$$

Work done:

$$W = \vec{F} \cdot \Delta\vec{r} = (6,2) \cdot (16.5,1.5)$$

$$W = 6(16.5) + 2(1.5) = 99 + 3 = 102 \text{ J}$$

Thus,

$$\boxed{W(0 \to 3 \text{ s}) = 102 \text{ J}}$$

Work-energy theorem check

Initial kinetic energy:

$$K_i = \frac{1}{2}m|\vec{v}(0)|^2 = \frac{1}{2}(2)(1^2 + (-1)^2) = 2 \text{ J}$$

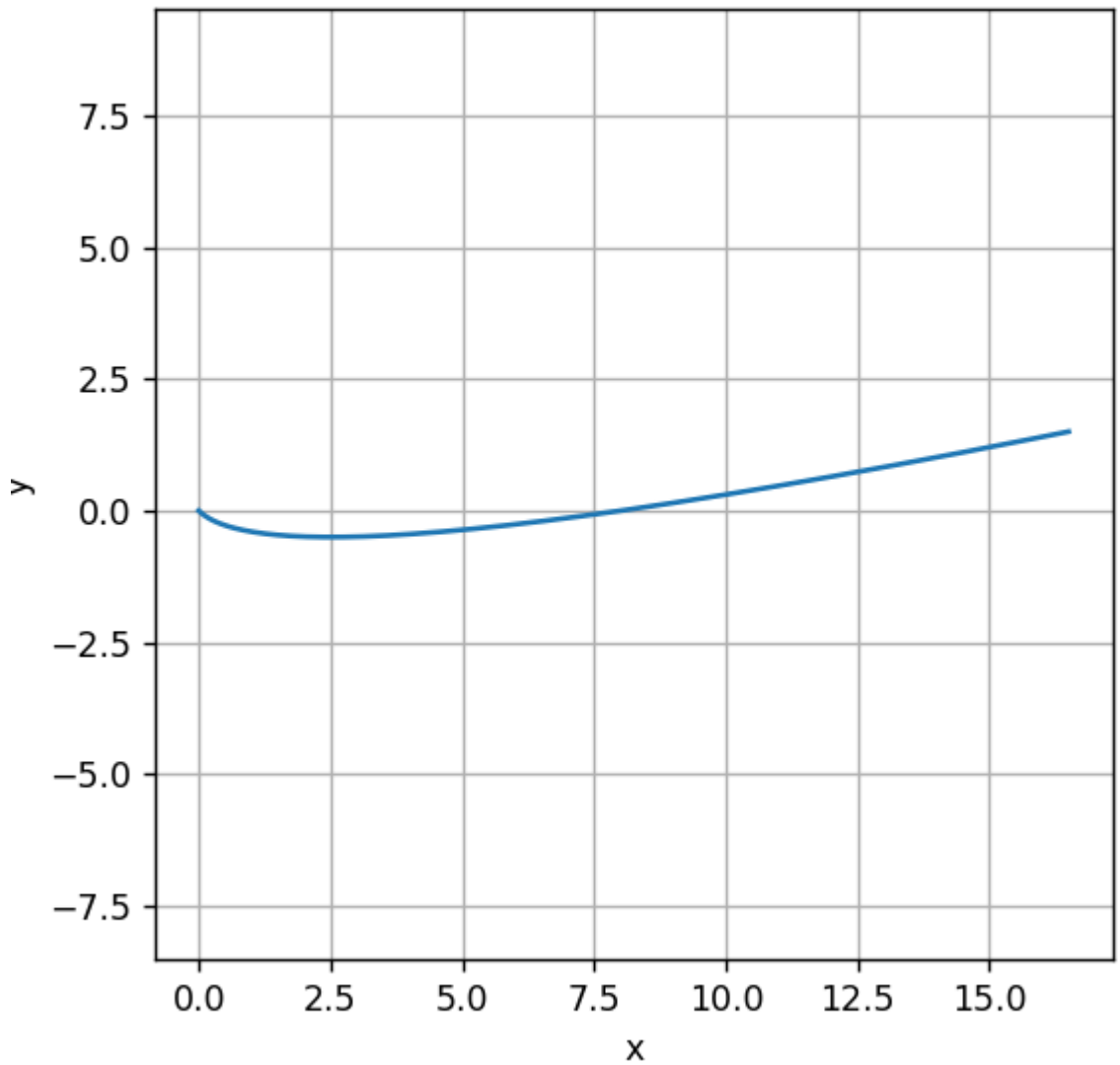
At $t = 3$,

$$\vec{v}(3) = (10,2)$$

so

$$K_f = \frac{1}{2}(2)(10^2 + 2^2) = 104 \text{ J}$$

Hence,



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$$\Delta K = K_f - K_i = 104 - 2 = 102 \text{ J}$$

which agrees with the work found above.

$$\boxed{W = \Delta K}$$